

SULAV SAPKOTA

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SUMMARY

Computer Engineering graduate specializing in deep learning, computer vision, and multilingual NLP. Built and deployed end-to-end low-resource AI systems for Nepali, spanning data pipelines, cloud GPU training, and production REST API deployment. Presented research on transformer-based Nepali language modeling at a national conference. Eager to apply research-backed ML skills to real-world AI/ML projects and grow within a fast-moving team.

EDUCATION

Bachelor of Engineering in Computer Engineering

2022 – 2026

Khwopa College of Engineering, Bhaktapur, Nepal

Relevant Courses: Data Structures and Algorithms, DBMS, Software Engineering, Computer Networks, Internet & Intranet

PROJECTS

Nepali Video Captioning Using Transformers

2024 – 2026

Bachelor Thesis • Presented at KHWOPA CEEL 2026 National Conference

- Built an end-to-end pipeline combining TimeSformer, Q-Former, and NLLB-200-distilled-600M to generate video captions in Devanagari script; applied LoRA fine-tuning on MSVD-Nepali and VATEX datasets.
- Achieved BLEU-4 34.07, METEOR 58.24, ROUGE-L 64.53, CIDEr 79.74; evaluation via SacreBLEU with intl tokenization and NFC normalization for Devanagari.
- Trained on cloud GPUs (Kaggle, RunPod) with HuggingFace Hub for checkpoint storage; deployed via FastAPI backend with a custom HTML/CSS/JS demo frontend.

Credit Card Fraud Detection

2024

- Benchmarked Random Forest, XGBoost, CatBoost, and Logistic Regression on a severely imbalanced dataset (0.17% fraud, 284K transactions); used SMOTE for linear models and scale_pos_weight for tree-based models.
- Tuned decision thresholds per model (XGBoost: 0.30, CatBoost: 0.60); used PR-AUC over ROC-AUC as primary metric. Best: Random Forest with 91.32% recall, 93.53% precision, F1 92.40%.

ACHIEVEMENTS

1st Place — Wine Quality Classification Competition

July–August 2024

Dept. of Computer & Electronics Engineering, Khwopa College of Engineering

CERTIFICATES

Deep Learning with Python: Foundation and Application (45 hrs)

Jan–Feb 2025

KhCE, CECAB, Almonds AI & Sciever Inc. • Computer Vision specialization — Excellence performance

Data Science with Python

Jun–Aug 2024

KhCE & DeepMind Creations Pvt. Ltd. • ML algorithms, data analysis, model evaluation — Excellent performance

Paper Presentation — Video Captioning in Nepali Using Transformers

Feb 2026

KHWOPA CEEL 2026 — National Conference on Computer, Electrical, Electronics & Communication

SKILLS

Languages: Python, C, C++, JavaScript

ML / DL Frameworks: PyTorch, Scikit-learn, HuggingFace Transformers, PEFT (LoRA)

Backend & Databases: FastAPI, Node.js, Express.js, MongoDB, SQL

Domains: Computer Vision, Natural Language Processing

Data & Analysis: NumPy, Pandas, Matplotlib, Seaborn

Tools: Git, GitHub, HuggingFace Hub, Kaggle, RunPod